Questions

Q1.

An athlete uses a training machine in a gym.

The display on the machine shows the time spent on the machine and the amount of energy transferred during a training session.

Figure 5 shows the displays for two different sessions by the same athlete.

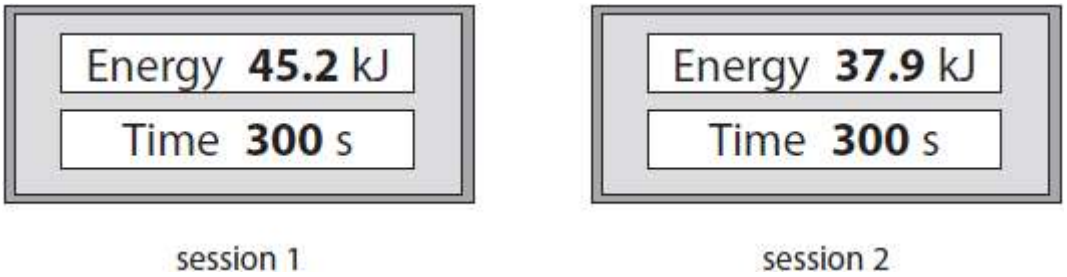

Figure 5

Explain what the displays show about the average power of the athlete in each of these two sessions.

(2)

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.....

(Total for question = 2 marks)

Q2.

A cyclist is riding a bicycle at a steady velocity of 12 m/s.

The cyclist and bicycle have a total mass of 68 kg.

The cyclist starts to cycle again.

The cyclist does 1600 J of useful work to travel 28 m.

Calculate the average force the cyclist exerts.

(3)

average force = ... N

(Total for question = 3 marks)

Q3.

A rock falls off the top of a cliff of height *h*.

Figure 3 shows the rock falling.

The Earth exerts a force of 150 N on the rock.

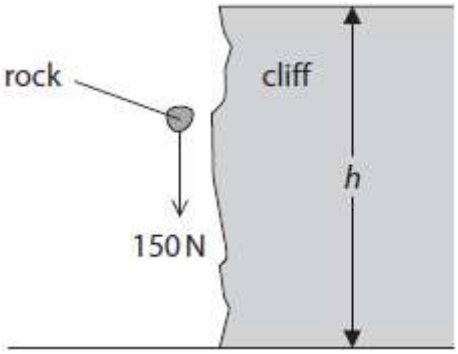

Figure 3

The work done by this force when the rock falls from the top to the bottom of the cliff is 2700 J.

(i) Calculate the height, *h*, of the cliff.

(2)

h = ... m

(ii) State the value of the kinetic energy of the rock just before it hits the ground.

(1)

kinetic energy = ... J

(iii) The mass of the rock in Figure 3 is 15 kg.

Calculate the velocity of the rock just before it reaches the bottom of the cliff.

(2)

velocity = ... m/s

(Total for question = 5 marks)

Q4.

Figure 22 shows a storage heater.

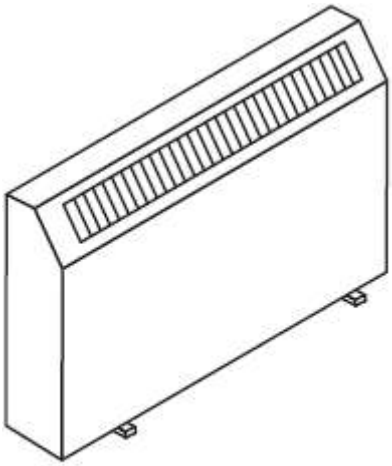

Figure 22

The storage heater contains bricks.

The bricks are heated electrically.

The electrical heater supplies 210 kJ of energy to each brick in the storage heater.

One brick has a mass of 5.8 kg.

The specific heat capacity for the brick is 860 J/kg K.

(i) Use this data to calculate the increase in temperature of the brick.

(2)

temperature increase = ... °C

(ii) The actual temperature increase will be smaller than you calculated in (i).

Explain why the actual temperature increase will be smaller than the value in (i).

(2)

.....

.....

.....

.....

(Total for question = 4 marks)

Q5.

A ball has a mass of 0.046 kg.

(i) Calculate the change in gravitational potential energy when the ball is lifted through a vertical height of 2.05 m.

Use the equation

$$\Delta GPE = m \times g \times \Delta h$$

(2)

change in gravitational potential energy = ... J

(ii) The ball is released.

Calculate the kinetic energy of the ball when the speed of the ball is 3.5 m/s.

(3)

kinetic energy of the ball = ... J

(iii) The ball bounces several times.

Figure 4 shows how the height of the ball above the floor changes with time.

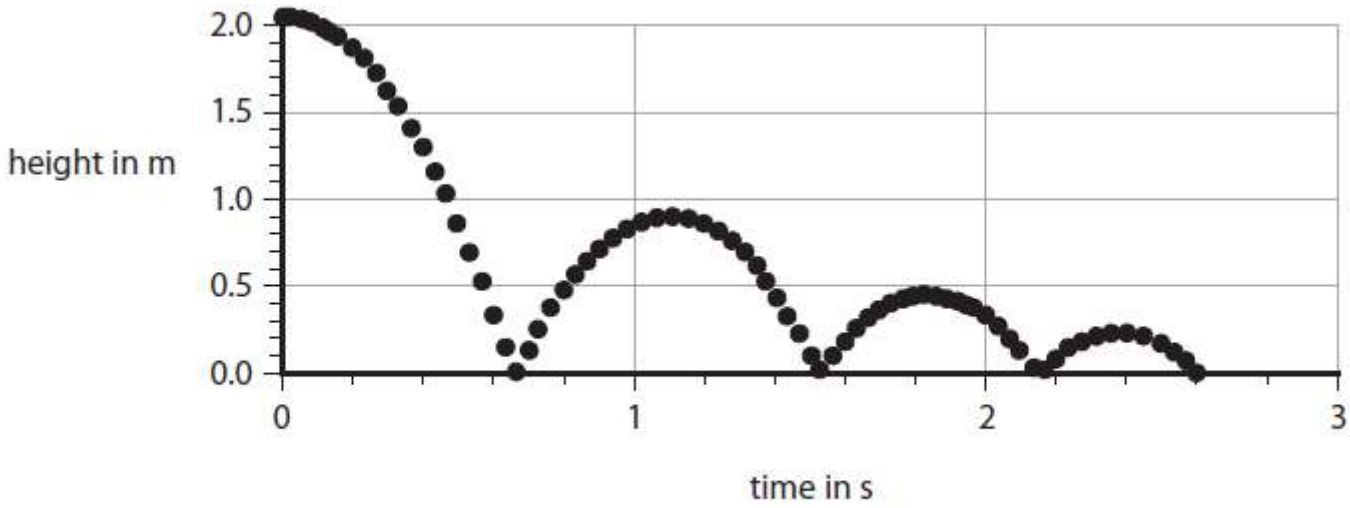

Figure 4

Use Figure 4 to estimate the maximum height that the ball reaches after the first bounce.

(1)

height after first bounce = ... m

(iv) Explain why the ball does not bounce back to its starting height of 2.05 m.

(2)

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.....

.....

.....

(Total for question = 8 marks)

Q6.

A different student finds out how much power can be generated by running up some stairs in a measured amount of time.

The following quantities are measured:

mass of student = 48 kg
total height of stairs = 2.9 m
time to run up stairs = 3.4 s

Calculate the power the student generated.

Give the answer to 2 significant figures (2 s.f.).

Use the equation

$$\text{power} = \frac{\text{weight of student} \times \text{height of stairs}}{\text{time to run up stairs}}$$

(3)

power = ... W

(Total for question = 3 marks)

Q7.

A drone has a mass of 4.5 kg.

This drone rises from the ground to a height of 20 m.

(i) Calculate the change in gravitational potential energy when the drone rises through a height of 20 m.
The gravitational field strength g = 10N/kg.

(2)

change in gravitational potential energy = ... J

(ii) State the amount of useful work done by the blades as the drone rises through 20 m.

(1)

useful work done = ... J

(iii) It takes 4 s for the drone to rise through 20 m.
Calculate the useful power developed by the blades in this time of 4 s.

(2)

useful power developed = ... W

(Total for question = 5 marks)

Q8.

Figure 15 shows a 'Mars rover' descending to the surface of the planet Mars.

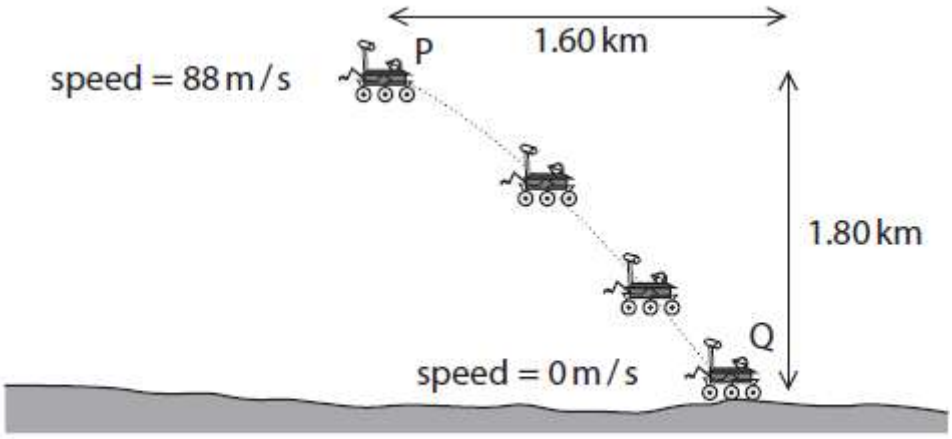

Figure 15

(i) Calculate the change in gravitational potential energy of the rover as it descends from position P to position Q.

Mass of rover = 1100 kg

Gravitational field strength on Mars = 3.7 N / kg

Give your answer to 2 significant figures.

(3)

change in gravitational potential energy = ... J

(ii) Use data from Figure 15 to calculate the change in kinetic energy of the rover as it descends from position P to position Q.

(2)

change in kinetic energy = ... J

(iii) The rover is slowed down safely using thrusters and a parachute (not shown in Figure 15).

The thrusters use jets of gas to control movements and the parachute is designed to be used in the atmosphere of Mars.

Describe the energy changes involved in terms of the work done by various forces as the rover descends.

(3)

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.....

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.....

(Total for question = 8 marks)

Q9.

Figure 17 shows an athlete training with a push sled.

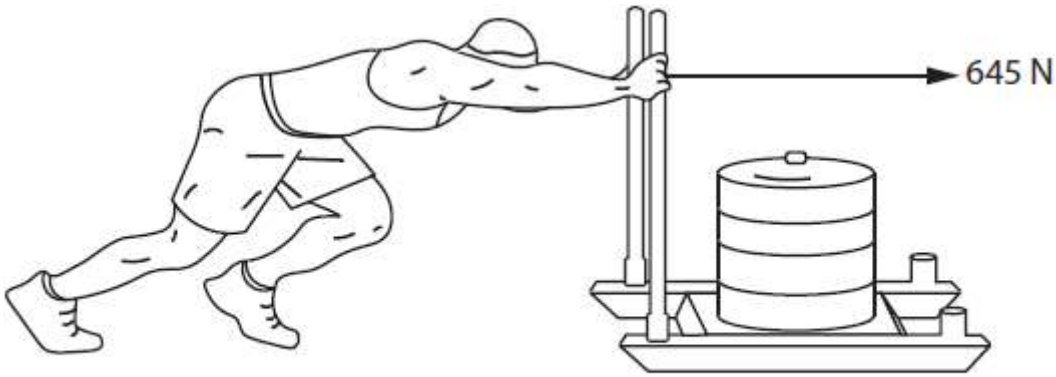

Figure 17

The athlete pushes the sled with a force of 645 N.

Calculate the distance the sled moves when the force of 645 N does 7440 J of work on the sled.

Give your answer to an appropriate number of significant figures.

(3)

distance moved = ... m

(Total for question = 3 marks)

Q10.

An electric car is travelling at a speed of 16.0 m/s.

The total mass of the car is 1200 kg.

(i) Calculate the kinetic energy, in kJ, of the car.

(2)

kinetic energy = ... kJ

(ii) On a journey, the car transfers energy from the battery at an average rate of 17.5 kW.

The battery in the car transfers a total of 126 MJ of energy before it becomes discharged.
Calculate the time taken for the battery to become discharged on this journey.
Give your answer in hours.

(2)

time taken = ... hours

(iii) Figure 20 shows an electrical device connected to the wheels of an electric car.

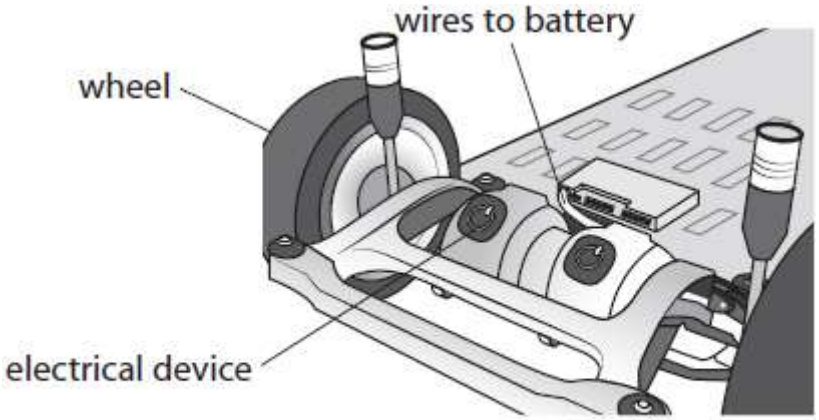

Figure 20

The electrical device is used as a motor when the car accelerates and as a dynamo when the car decelerates.
Explain how using the device can help to increase the time that the car can be driven before the

battery becomes discharged.

(2)

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(Total for question = 6 marks)

Q11.

Figure 8 shows a bicycle.

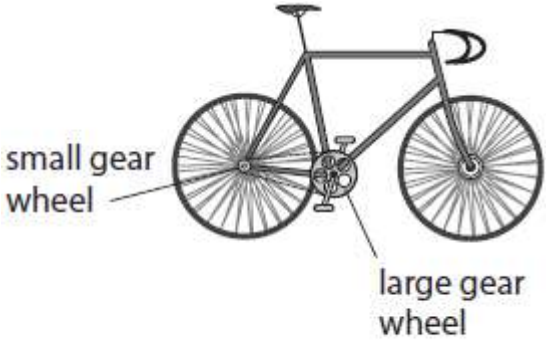

Figure 8

(i) The rider uses the pedals to make the large gear wheel turn.

The large gear wheel moves the chain.
The chain turns the small gear wheel.
The large gear wheel has 48 teeth.
The small gear wheel has 12 teeth.
The large gear wheel turns 2 times each second.
Calculate the number of times that the small gear wheel turns each second.

(2)

..... turns each second

(ii) Oil is applied to the wheel of a bicycle at the point shown in Figure 9.

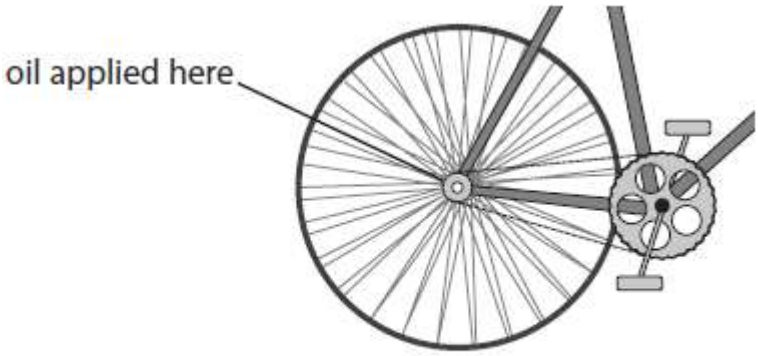

Figure 9

Explain how the oil improves the efficiency of the bicycle.

(3)

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.....

(Total for question = 5 marks)

Mark Scheme

Q1.

Question Number:	Answer	Additional Guidance	Mark
	an explanation linking:	allow reverse argument	**(2)** AO 3 2a AO 3 2b
	over the **same time / in 300s**, more work done / energy transferred in session 1 than in session 2 (1)	power in session 1 = $\frac{45.2}{300}$ = 0.15 (kW) or 150(W) allow statement that power = $\frac{\text{work}}{\text{time}}$ / or power = $\frac{\text{energy(transferred)}}{\text{time}}$ for MP1	
	(therefore) more power (developed) in session 1 (1)	power in session 2 = $\frac{37.9}{300}$ = 0.13 (kW) or 126(W)	

Q2.

Question Number:	Answer	Additional Guidance	Mark
	recall and substitution (1) 1600 = force x 28 rearrangement (1) (force) = $\frac{1600}{28}$ evaluation (1) 57 (N)	substitution and rearrangement in either order accept f, F or ? for force accept values that round down to 57 e.g. 57.14 award full marks for correct answer without working award 1 mark for answers of 44800 or 0.0175 **and** a correct expression relating work, force and distance	**(3)** AO 2 1

Q3.

Question number	Answer	Additional guidance	Mark
(i)	rearrangement of work = force × distance to give distance = work ÷ force (1) substitution and evaluation (1) 18 (m)	seeing 2700 ÷ 150 Award full marks for correct answer without working	**(2)**

Question number	Answer	Mark
(ii)	2700 (J)	**(1)**

Question number	Answer	Additional guidance	Mark
(iii)	rearrangement of KE = ½ mv^2 $v = \sqrt{(2 \times KE \div m)}$ (1) substitution and evaluation (1) 19 (m/s)	 $v = \sqrt{(2 \times 2700 \div 15)}$ $v^2 = (2 \times 2700 \div 15)$ allow answers that round to 19 award full marks for correct answer without working allow alternative route using $v^2 - u^2 = 2ax$ for full marks	(2)

Q4.

Question number	Answer	Additional guidance	Mark
i	rearrangement **and** substitution (1) $(\Delta\theta =) \frac{210 (x10^3)}{5.8 \times 860}$ evaluation (1) 42 (°C)	 $(\Delta\theta =) \frac{210 (x10^3)}{4988}$ accept any value which rounds to 42 e.g. 42.10 award full marks for the correct answer without working 4.2 to any other power of 10 scores 1 mark	**(2) AO2.1**

Question number	Answer	Additional guidance	Mark
ii	an explanation linking any **two** from	ignore: • energy is lost / wasted, unqualified • not 100% efficient • arguments about sound energy accept heat for energy throughout	**(2)** **AO2.1**
	not **all** the energy supplied goes to the brick (1)	less (thermal) energy given to brick	
	not all the energy supplied stays in the brick (1)	energy transfers from the brick	
	energy transferred to the storage heater parts (1)		

	energy transferred to the surroundings (1)	energy dissipated	
	argument linking Δθ to ΔQ **using** $\Delta\theta = \frac{\Delta Q}{m \times c}$ (1)	from the equation, if energy supplied to the block is smaller the change of temperature will be smaller 'brick transfers (thermal) energy to the surroundings' scores 2 marks	

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	substitution (1) (ΔGPE=) (0.0)46×10×2.05 evaluation (1) 0.94(3) (J)	allow g=9.8(1) m/s^2 0.9 (J) values that round to 0.92 or 0.93 (from using g = 9.8 or 9.81) do not award for 1(J) no POT error in evaluation award full marks for the correct answer without working.	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	recall (1) (KE=) $\frac{1}{2} \times m \times v^2$ substitution (1) (KE=) $\frac{1}{2} \times (0.0)46 \times 3.5^2$ evaluation (1) 0.28 (J)	allow answers that round to 0.28 e.g. 0.28175 (J) allow max 2 marks for POT error e.g. 0.00028 award full marks for the correct answer without working	**(3)**

Question Number	Answer	Additional guidance	Mark
(iii)	Any value between 0.8 (m) and 0.95 (m) inclusive		**(1)**

Question Number	Answer	Additional guidance	Mark
(iv)	An explanation linking (the ball) has lost energy (1) identification of what has happened to that energy (1)	accept (energy) dissipated **or** (transferred to) surroundings / ground **or** thermal energy **or** heat / sound **or** system is not 100% efficient **or** bounce is not (100%) elastic **or** squashing (the ball or the ground)	**(2)**

Q6.

Question Number	Answer	Additional guidance	Mark
		accept 9.8(1) instead of 10 for g throughout	**(3)** **AO 2.1**
	substitution (1) (P =) $\frac{48 \times 10 \times 2.9}{3.4}$	$\frac{480 \times 2.9}{3.4}$ OR $\frac{1392}{3.4}$	
	evaluation (1) 409 (W)	accept values that round to 409 accept values that round to 402 (used g = 9.81)	
	to 2 s.f. (1) 410 (W)	accept the result of any **calculation** to 2s.f. for this mark accept 400 (W) if 402 seen (used g = 9.81) award full marks for the correct answer without working	

Q7.

Question number	Answer	Additional guidance	Mark
(i)	recall and substitution into (1) gpe = m x g x h (gpe) = 4.5 x 10 x 20 evaluation (1) 900(J)	 allow 90(J) for 1 mark award full marks for the correct answer without working	(2)

Question number	Answer	Additional guidance	Mark
ii	900(J)	allow ecf from i	(1)

Question number	Answer	Additional guidance	Mark
iii	recall and substitution (1) power = work done / time taken (power =) 900 / 4 evaluation (1) 200 (W)	allow ecf from i or ii accept 230(W), 225(W) award full marks for the correct answer without working	(2)

Q8.

Question number	Answer	Additional guidance	Mark
(i)	select and substitute (1) ($\Delta GPE = m \times g \times \Delta h$) $= 1100 \times 3.7 \times 1.8 (\times 10^3)$ evaluation (1) 7 326 000 (J) evaluation to 2 s.f. (1) 7 300 000 (J)	any number rounding to 7 300 000 7326 scores 1 mark **independent mark** - any final answer stated to 2 s.f.	**(3)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	select and substitute (1) ($\Delta KE = ½ m \times v^2$) $= ½ 1100 \times 88^2$ evaluation (1) 4 300 000 (J)	ignore minus signs accept numbers that round to 4 300 000 (J) e.g. 4 259 200 (J) award full marks for the correct answer without working	**(2)** **AO2.1**

Question number	Answer	Additional guidance	Mark
iii	A description linking **three** from: 1. work is done against / by gravity (1) 2. **idea of** work done by the thrusters / jets (on the rover) (1) 3. (work done) by air/atmospheric resistance on the parachute (and rover) (1) 4. this reduces the kinetic energy (store) (1) 5. (there is a) decrease in the gravitational potential energy (store) of the rover (1) 6. (there is a) transfer of chemical energy from the thrusters (1) 7. energy transferred to thermal energy (store) (1) 8. (transfer) mechanically (to the thermal store) (1)	**KEY: attempt to explain <u>how work done</u> contributes towards the energy changes / conservation of energy** if no other mark scored allow one mark for work = force x distance	**(3) AO2.1**

Q9.

Question number	Answer	Additional guidance	Mark
		rearrangement and substitution in either order	(3) AO2.1
	substitution (1) 7440 = 645 x distance	allow substitution of correct values into a visible, incorrectly rearranged algebraic equation for this mark only	
	rearrangement (1) (distance =) $\frac{7440}{645}$	(distance =) $\frac{\text{work done}}{\text{force}}$	
	evaluation and rounding to 3sf (1) 11.5 (m)	11.53 (m) scores 2 marks only award full marks for the correct answer without working	

Q10.

Question number	Answer	Additional guidance	Mark
(i)	selection and substitution (1) (KE =) ½ x 1200 x 16(.0)^2 evaluation in kJ (1) (KE =) 150 (kJ)	(KE =) ½ x 1200 x 16(.0)^2 x 10^{-3} accept any value that rounds to 150 e.g. 153.6 award full marks for correct answer without working. award 1 mark for 153.6 or 150 to any other power of ten	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	selection and substitution (1) $17.5\ (\times 10^3) = \frac{126\ (\times 10^6)}{t}$ re-arrangement and evaluation (1) (t=) 2(.0) (h)	alternative method selection and rearrangement (1) $(t =)\ \frac{E(nergy)}{P(ower)}$ or $(t=)\ \frac{126\ (\times 10^6)}{17.5\ (\times 10^3)}$ (substitution and) evaluation (1) (t=) 2(.0) (h) award full marks for correct answer without working. allow 1 mark for 7(.2) to any power of ten (incorrect time conversion) allow 1 mark for 2(.0) to any power of 10 (POT error)	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	an explanation linking (energy transfers when the car is decelerating) (from) kinetic energy (store) (1) (to) chemical energy (store) (1)	 idea of energy that would be otherwise wasted uses an electrical pathway {electric current / electricity / emf} produced allow mechanical for kinetic in this context recharges battery increases available energy store of battery more useful energy available	**(2)** **AO2**

Q11.

Question Number	Answer	Additional guidance	Mark
(i)	(In every second), distance moved by chain around large gear = distance moved by chain around small gear (1) 2 x 48 = turns x 12 rearrangement and evaluation (1) 8 (turns each second)	accept use of gear ratio seen or implied e.g. 4:1 or 4/1 or 48:12 or 48/12 or converse e.g. 1:4 award full marks for the correct answer without working	**(2)**

Question Number	Answer	Additional guidance	Mark
(ii)	An explanation linking reduces friction/amount of thermal energy transferred (1) extra useful energy is available/less input energy is required (1) efficiency = useful energy transferred (by the bicycle) ÷ total energy supplied (to the bicycle) (1)	 (oil provides) lubrication less energy wasted allow for the last **two** mark points; **either** less input energy is required to produce the same output for **2** marks **or** more output energy is available for the same input energy for **2** marks	**(3)**